

Ming-Qiang Liang

Ph.D. Candidate in Climate Dynamics

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Research Interests

Paleoclimate modeling; Asian monsoon dynamics; AMOC variability; proxy–model comparison; Heinrich events.

Education

Ph.D. Climate Dynamics, UCLouvain (Advisor: Prof. Qiuzhen Yin)	2026
M.S. Geoscience, National Taiwan University (Advisor: Prof. Hongchun Li)	2021
B.S. Geoscience, Southwest University (Advisor: Prof. Tingyong Li)	2018

Awards

FNRS Aspirant Fellowship	2022–2026
National Taiwan University Graduate Scholarship	2019–2020
Outstanding Graduate, Southwest University	2018

Publications

In Press

[1] **Liang, M.**, Yin, Q. (in press). CO₂ forcing drives divergent tropical hydroclimate during Heinrich stadials.

2026

[2] Sun, W., Tao, Z., Zhang, Z., Lan, T., Li, T., Zi, P., Zhang, J., Wu, Y., **Liang, M.**, Chen, C., Mu, Y., Li, J., Li, T. (2026). Coupled influence of Western Pacific moisture anomalies and mid-tropospheric humidity on October precipitation in Yunnan Province Southwest China. *Atmospheric Research*, 336, 108830. <https://doi.org/10.1016/j.atmosres.2026.108830>

2025

[3] Chen, C., Dong, Y., Wu, Y., Zhang, J., Wang, H., Xue, Y., Sun, W., Xie, M., Cheng, H., Li, J., **Liang, M.**^{*}, Li, T. (2025). The variation of the East Asian summer monsoon on multi-centennial scale during 8.3–2.9 ka BP. *Global and Planetary Change*, 255, 105115. <https://doi.org/10.1016/j.gloplacha.2025.105115>

[4] Zhang, J., **Liang, M.**, Li, J., Wu, Y., Xue, Y., Sun, W., Liu, X., Chen, J., Chen, C., Li, T. (2025). Upstream moisture sources and circulation patterns controls Asian monsoon hydroclimate over the past 20,000 years. *Quaternary Science Reviews*, 363, 109438. <https://doi.org/10.1016/j.quascirev.2025.109438>

[5] Zhao, J., Cheng, H., Pan, L., Wang, K., Cui, J., **Liang, M.**, Dong, X., Lu, J., Li, Q., Yang, Y. (2025). Putting the recent 100-year of Asian summer monsoon weakening into context. *Science Bulletin*, 70(18), 2959–2962. <https://doi.org/10.1016/j.scib.2025.05.006>

[6] Liu, Y., Hu, C., Hu, Z., Liao, J., **Liang, M.**, Yin, Q. (2025). Extended duration of abrupt climate events from the early to late Holocene. *Geophysical Research Letters*, 52(7). <https://doi.org/10.1029/2025gl115543>

2024

[7] **Liang, M.**, Yin, Q., Sun, Y., Zhang, C., Wu, Z., Liu, W. (2024). Distinct response of Asian summer precipitation and monsoon circulation to orbital forcing during Heinrich events. *Quaternary Science Reviews*, 344, 108946. <https://doi.org/10.1016/j.quascirev.2024.108946>

2023

[8] Zhang, J., Genty, D., Sirieix, C., Verdet, C., Mateo, S., Devaux, L., Sgubin, G., Bujan, S., Minster, B., Régnier, E., Li, T., Bourges, F., **Liang, M.**, Chen, C., Mariéu, V. (2023). Data-model comparisons of isotopic and hydrological variabilities of the karstic vadose zone above Villars Cave, SW-France based on 20 years' monitoring record. *Chemical Geology*, 642, 121802. <https://doi.org/10.1016/j.chemgeo.2023.121802>

[9] Xue, Y., Zhang, J., Su, Z., Wu, Y., Liang, Q., **Liang, M.**, Liu, X., Chen, C., Qiu, H., Li, J., Li, T. (2023). Quantifying source effects based on rainwater $\delta^{18}\text{O}$ from 10-year monitoring records in Southwest China. *Applied Geochemistry*, 155, 105706. <https://doi.org/10.1016/j.apgeochem.2023.105706>

[10] Wu, Y., Li, T., Li, J., Cheng, H., Ning, Y., Shen, C., Yang, Y., Zhao, J., Chen, C., **Liang, M.**, Xiao, S., Qiu, H., Xu, Y., Huang, Y., Yu, T., Edwards, R. (2023). Variations in Asian summer monsoon and hydroclimate during Heinrich stadials 4 revealed by stalagmite stable isotopes and trace elements. *Quaternary Science Reviews*, 299, 107869. <https://doi.org/10.1016/j.quascirev.2022.107869>

2022

[11] Wu, Z., Yin, Q., **Liang, M.**, Guo, Z., Shi, F., Lu, H., Su, Q., Lü, A. (2022). The effect of astronomical forcing on water cycle: Sea ice and precipitation. *Chinese Science Bulletin*, 68(12), 1443–1458. <https://doi.org/10.1360/tb-2022-0833>

[12] Zhang, J., **Liang, M.**, Li, T., Chen, C., Li, J. (2022). Asian-Australian monsoon evolution over the last millennium linked to ENSO in composite stalagmite $\delta^{18}\text{O}$ records. *Quaternary Science Reviews*, 281, 107420. <https://doi.org/10.1016/j.quascirev.2022.107420>

2021

[13] Chen, C., Yuan, D., Li, J., Wang, X., Cheng, H., Ning, Y., Edwards, R., Wu, Y., Xiao, S., Xu, Y., Huang, Y., Qiu, H., Zhang, J., **Liang, M.**, Li, T. (2021). The 4.2 ka event in East Asian monsoon region, precisely reconstructed by multi-proxies of stalagmite. <https://doi.org/10.5194/cp-2021-20>

Conference Presentations

Oral Presentations

[1] Distinct responses of Asian summer monsoon to the Orbital forcing during the last glacial Heinrich events. *The 7th PAGES Open Science Meeting*. Shanghai, China. 2025

[2] Distinct responses of Asian summer monsoon to the Orbital forcing during the last glacial Heinrich events. *EGU*. Vienna, Austria. 2024

[3] Asynchronous Holocene Optimum in East Asia monsoon region recorded by stalagmites and its underlying climate dynamics. *XXI Congress of the International Union for Quaternary Research (INQUA)*. Rome, Italy. 2023

[4] Paleoclimatic evolutions during the Holocene: A stalagmite $\delta^{18}\text{O}$ record from Majiaping Cave, Guizhou, China. *EGU*. Vienna, Austria. 2022

[5] The highest resolution stalagmite over Holocene in southwest China by accurate ^{14}C dating. *Climate Change: The Karst Record IX*. Innsbruck, Austria. 2021

Poster Presentations

[1] Variations of ENSO linking with the intensity of Asian-Australian monsoon revealed by stalagmite and rainfall $\delta^{18}\text{O}$ over the past millennium. *The 28th General Assembly of the International Union of Geodesy and Geophysics*. Berlin, Germany. 2023

[2] Chronology construction of a stalagmite from southwest China by ^{14}C , $^{230}\text{Th}/\text{U}$ and ^{210}Pb dating methods. *The 8th East Asia Accelerator Mass Spectrometry Symposium*. Nagoya, Japan. 2019

Research Experience

Sun Yat-Sen University, Guangzhou, China	2024.10
Climate Feedback–Response Analysis Method (CFRAM)	
Xiamen University, Xiamen, China	2023.02
Isotope-enabled climate modeling	
Institute of Atmospheric Physics, CAS, Beijing, China	2022.12
Regional climate modeling (RegCM4)	

Technical Skills

Climate models: HadCM3, iCESM, RegCM4.7

Programming: Matlab, Fortran, Bash, Slurm

Climate Diagnostics: Data Processing & Analysis, Physical Mechanism Diagnostics, Climate Model Evaluation

Paleoclimate: Proxy, Dating, Uncertainty

Academic Service

Journal reviewer:

Quaternary Science Reviews; Climate of the Past; Climate Dynamics; Global and Planetary Change